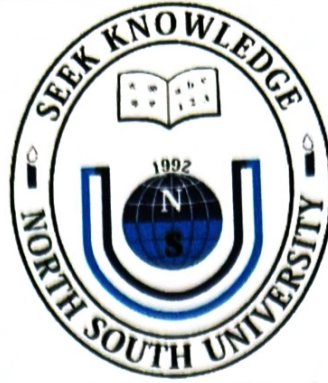




North South University
Department of Electrical & Computer Engineering

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Course Title: Analog Electronics I Lab

Faculty Name: Ms. Syeda Sarita Hassan [SSH1]

Section: 7

Experiment Number: 06

Experiment Name:

The BJT Biasing Circuits

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Group Number: 01

Submitted To: Md. Mahbub Hassan

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Experiment name: The BJT Biasing Circuits.

Objective: This experiment aims,

- to study different BJT biasing circuits.
- to understand the purpose of biasing in BJT.

Theory: Biasing a BJT circuit means giving the transistor the correct DC voltages and currents from external sources so that it has a proper operating point, called the Q-point, in the active region.

After the Q-point is set, the input signal should cause an output signal that has the same shape.

If the output doesn't match the input, for example, if part of the waveform is clipped, then the Q-point is not correct and must be moved to a better position on the collector characteristics.

So, the main goal of biasing a BJT is to choose the right Q-point so the output signal can reproduce the input signal faithfully. There are several types of biasing circuits. But we will mainly focus on fixed-

bias and self-bias circuits.

In the fixed-bias circuit, the base current I_B is set by the base resistor R_B and stays constant.

The major disadvantage of this method is that the Q-point becomes unstable when the transistor's β value changes.

In the self-bias circuit, this problem is solved by using the emitter resistor R_E , which provides self-biasing and improves stability.

Equipment List:

Serial no.	Component Details	Specification	Quantity
1.	Transistor	C828, BD135	1 piece each
2.	Resistor	470 Ω , 2.2k Ω , 3.3k Ω , 4.7k Ω , 10k Ω , 470k Ω	1 piece each
3.	POT	10k Ω , 100k Ω	1 piece each
4.	Trainer Board		1 unit
5.	DC Power Supply		1 unit
6.	Digital Multimeter		1 unit
7.	Chords and wires		as required

Circuit Diagram :

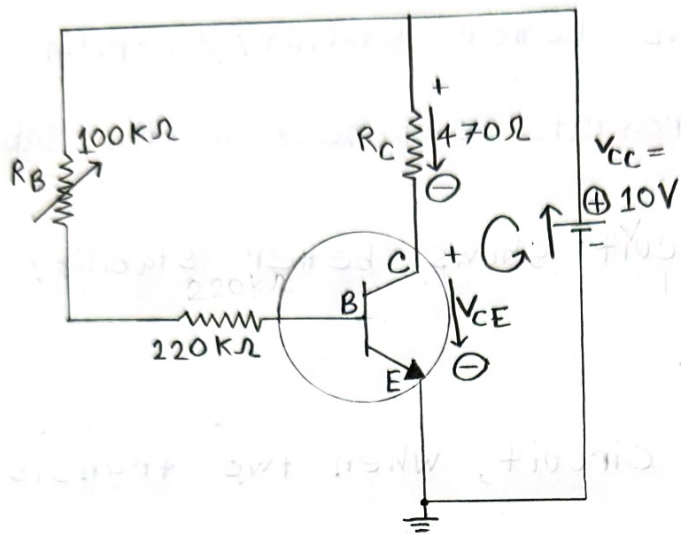


Fig 1: Fixed Bias Circuit

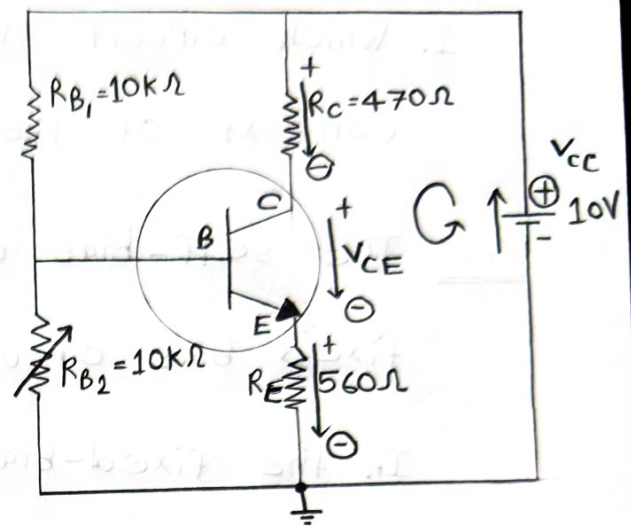


Fig 2: Self Bias Circuit

Data Table :

Table 1: Data for fixed bias circuit.

Transistor	$R_C (\Omega)$	V_C (Volt)	$I_C = V_C / R_C$ (Amp)	V_{CE} (Volt)	Q-Point
CB 28	460.8	5.040	0.011	5	(5, 0.011)
BD135	460.8	4.89	0.011	5.14	(5.14, 0.011)

Table 2: Data for self bias circuit.

Transistor	$R_C (\Omega)$	V_C (Volt)	$I_C = V_C / R_C$ (Amp)	V_{CE} (Volt)	Q-Point
CB 28	460.8	2.24	0.005	5.05	(5.05, 0.005)
BD135	460.8	1.72	0.004	4.99	(4.99, 0.004)

Question and Answer :

1. Which circuit shows better stability? Explain in the context of the results obtained in the laboratory.

Ans: The self-bias circuit shows better stability than the fixed bias circuit.

In the fixed-bias circuit, when two transistors (C828, BD-135) with different β were used, the Q-point shifted, especially the V_{CE} value. From table 1, we can see that, Q-point shifted from (5, 0.011) to (5.14, 0.011). This happens because, in fixed bias, the base current I_B is set only by R_B . So, any change in β directly changes I_C ; causing the Q-point to become unstable.

In the self-bias circuit, the Q-points for both transistors remained almost the same. From table 2, we can see that, Q-point shifted from (5.05, 0.005) to (4.99, 0.004). Here, V_{CE} remained around 5V and I_C slightly changed. This stability is due to the emitter resistor R_E ; which provides negative

feedback, meaning, if I_C increases, V_{BE} automatically decreases, controlling I_C and keeping the Q-point steady. The Q-points are very close, meaning the self-bias circuit is far more stable.

So, from theoretical explanation and from the results obtained in the laboratory, we can say that, self-bias circuit provides much better stability than the fixed-bias circuit.

2. Draw the DC load line for both the circuits and show the Q-point.

Ans: For fixed bias circuit:

By KVL in the fixed bias circuit (Circuit Diagram; Fig1),

$$+V_{CC} - V_{RC} - V_{CE} = 0$$

$$+V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$\begin{aligned} \text{When } I_C = 0 \text{ mA; } V_{CE} &= V_{CC} - 0 R_C \\ &= V_{CC} \\ &= 10 \text{ V} \end{aligned}$$

$$\text{When } V_{CE} = 0 \text{ V; } 0 = V_{CC} - I_C R_C$$

$$\begin{aligned} I_C &= \frac{V_{CC}}{R_C} = \frac{10}{470} \text{ A} \\ &= 21.28 \text{ mA} \end{aligned}$$

So, connecting the points (10,0) and (0, 21.28) gives the load line for fixed bias circuit.

And, from the data table, we can see that, for C828 transistor, the Q-point is (5, 11) or, (5V, 11mA) and for BD-135 transistor, the Q-point is (5.14, 11) or, (5.14V, 11 mA).

Graph of DC load line for fixed bias circuit and Q-points for both the transistors is attached.

For self bias circuit:

By KVL in the self bias circuit (Circuit Diagram; Fig 2),

$$+V_{CC} - V_{RC} - V_{CE} - V_{RE} = 0$$

$$V_{CE} = V_{CC} - V_{RC} - V_{RE}$$

$$= V_{CC} - I_C R_C - I_C R_E$$

$$= V_{CC} - I_C (R_C + R_E)$$

$$\begin{aligned} \text{When } I_C = 0 \text{ mA, } V_{CE} &= V_{CC} - 0(R_C + R_E) \\ &= V_{CC} \\ &= 10 \text{ V} \end{aligned}$$

$$\text{When } V_{CE} = 0 \text{ V; } 0 = V_{CC} - I_C (R_C + R_E)$$

$$\begin{aligned} I_C &= \frac{V_{CC}}{R_C + R_E} = \frac{10}{470 + 560} \text{ A} \\ &= 9.71 \text{ mA} \end{aligned}$$

So, connecting the points $(10,0)$ and $(0,9.71)$ gives the load line for self bias circuit,

From the data table, we can see that, for C828 transistor, the Q-point is $(5.05, 5)$ or, $(5.05\text{ V}, 5\text{ mA})$ and for BD-135 transistor, the Q-point is $(4.99, 4)$ or, $(4.99\text{ V}, 4\text{ mA})$.

Graph of DC load line and Q-points (both transistor) for self bias circuit is attached.

Discussion :

In this experiment, we studied two types of BJT biasing circuits (Fixed bias and self bias). The main goal was to see how stable the operating point (Q-point) is when different transistors are used.

For the fixed bias circuit, the Q-point changed when the transistor changed because fixed bias depends on the transistor's β value. When β changes, I_c and V_{ce} also change. So, the fixed bias circuit is not stable.

For the self bias circuit, the Q-point stayed almost the same for both transistors because the emitter resistor (R_E) provides negative feedback, which keeps I_C and V_{CE} steady even if β changes. So, self bias circuit is more stable.

Although the theory gave us the expected values, the real circuit showed small differences because of resistor tolerances and different β values. This shows real circuits are never exact; so we must consider these variations when designing.

During the experiment, we faced difficulties while using the transistors. However, the readings were stable throughout the experiment. In conclusion, the experiment shows that self bias gives a more stable Q-point than fixed bias. It also helped connect theory with real practice and gave us useful understanding about how BJT circuits work in real situations.

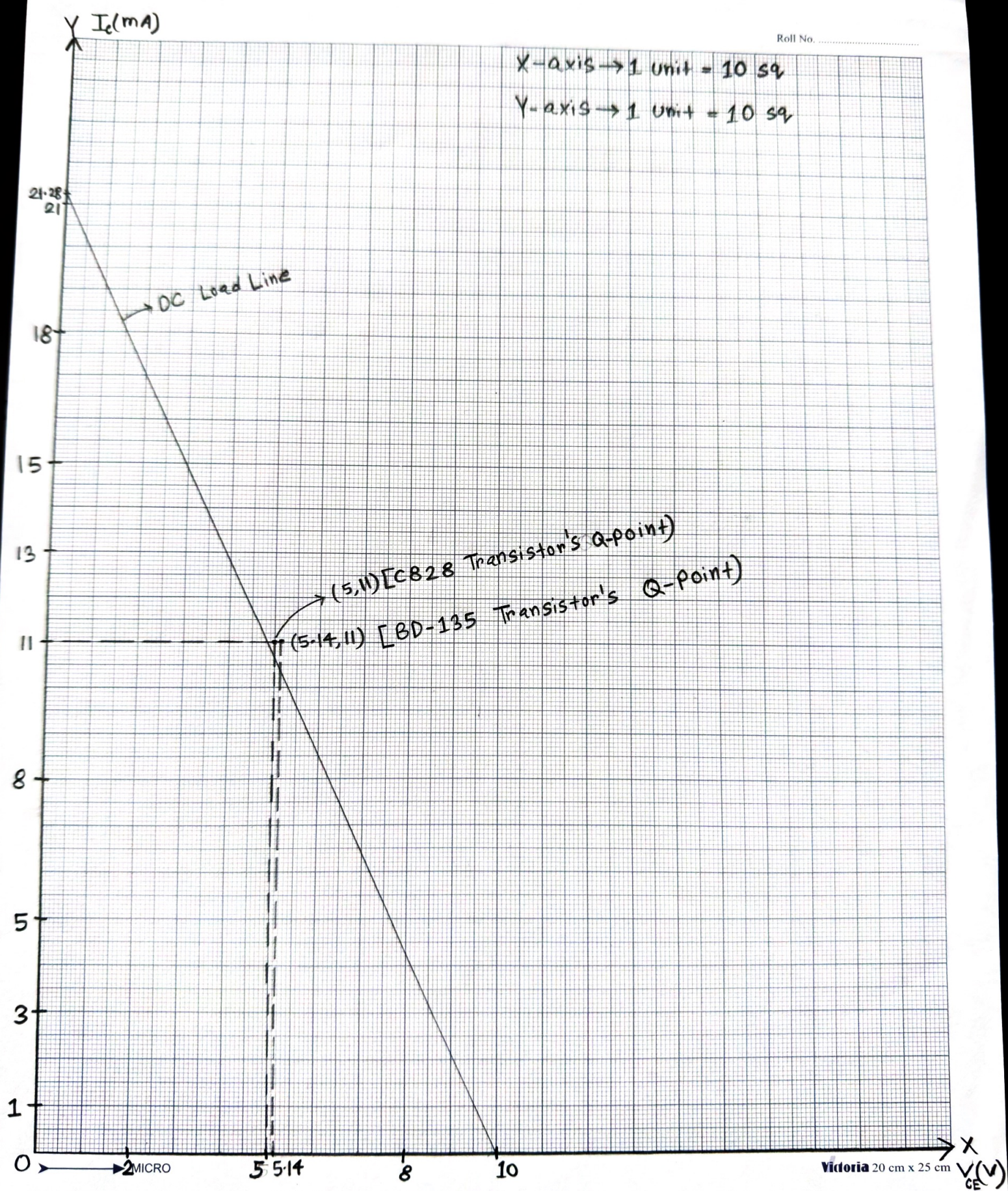


Fig: Fixed Bias Circuit

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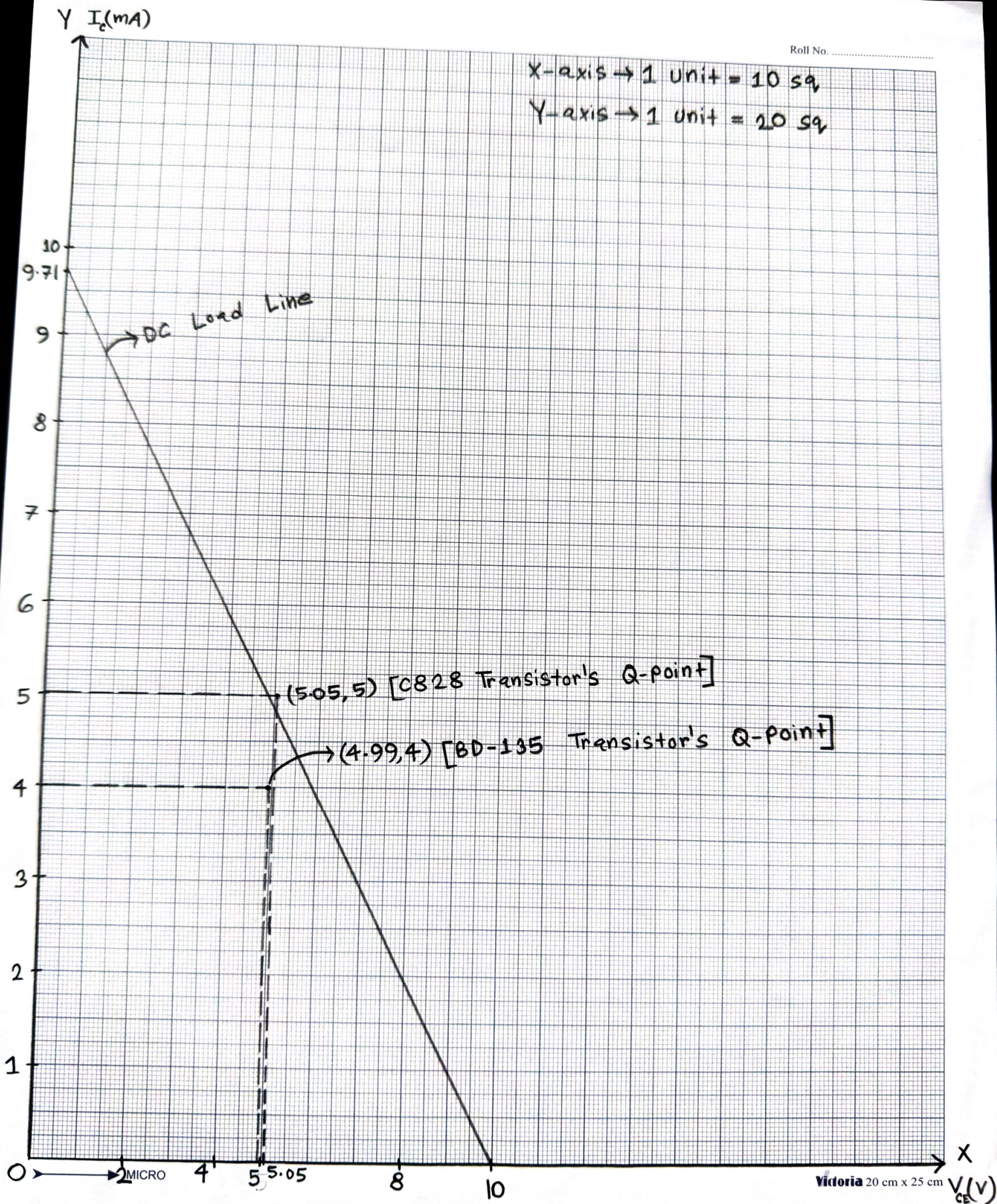


Fig: Self Bias Circuit